

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

LilPaws AI

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Transforming Pet Adoption Through AI



LilPaws is a modern, AI-powered pet adoption ecosystem designed to streamline the connection between rescued animals and loving homes. Our platform leverages Artificial Intelligence to pre-screen adoption applications, ensuring pets are matched with the most suitable environments.

By automating the initial screening process, we reduce administrative burden on shelters while improving matching accuracy and adoption success rates.

Smart Pet Discovery Experience



Intelligent Search

Browse and search for pets available for adoption with advanced filters



Digital Application

Comprehensive multi-step application process with validated criteria



Live Status Tracking

Monitor application progress from "AI Review" to "Final Decision" in real-time



Community Impact Features

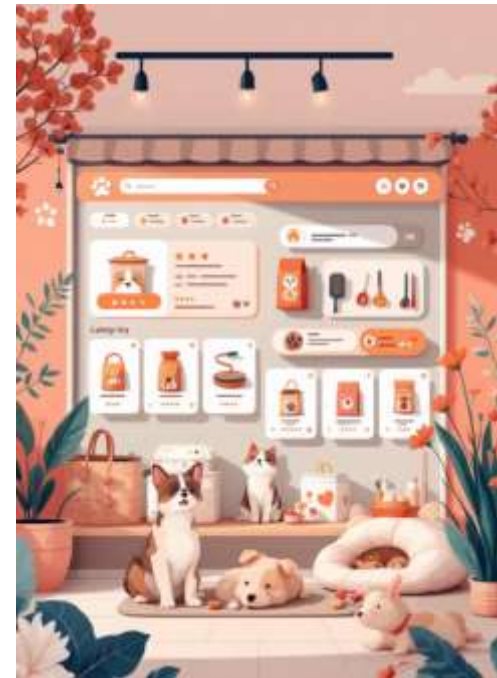
Report Strays

Community-driven reporting system helps animals in need find the nearest shelters quickly and efficiently.



E-commerce Hub

Comprehensive shop for pet supplies with integrated payments, supporting shelter operations.



AI-Assisted Screening for Shelters



1

Autonomous Mode

AI handles primary filtering and initial verdicts using Google Gemini

2

Manual Oversight

Admins review AI-flagged applications for final confirmation



Real-Time Control Center

Live-synced inbox for pending applications without page refreshes

01

Multi-Shelter Management

Dedicated controls for Chandigarh, Panchkula, and Mohali facilities

02

Flexible Assessment Modes

Choose between autonomous AI processing or manual review workflows

Microservices-inspired architecture with three distinct components working in harmony

1

Frontend Client

- React.js with Vite
- Tailwind CSS & Shadcn UI
- Redux Toolkit state management
- Lucide React icons

2

Primary Backend

- Node.js & Express
- MongoDB with Mongoose ODM
- Redis for cache/queue
- Cloudinary for images

3

AI Review Service

- Python FastAPI framework
- LangChain + Gemini Pro
- Asynchronous processing
- Motor async MongoDB driver

Technology Stack Highlights



Frontend Excellence

- React.js with Vite
- Tailwind CSS
- Shadcn UI components
- Framer Motion animations

Backend Power

- Node.js & Express
- MongoDB NoSQL
- Redis in-memory data
- PayPal SDK integration

AI Intelligence

- Python FastAPI
- Google Gemini Pro
- LangChain framework
- Asynchronous workers

Quick Start Guide

1

Prerequisites

- Node.js v18+
- Python 3.9+
- Redis Server
- MongoDB instance

2

Environment Setup

Configure .env files for server and AI service with MongoDB URI, Redis URL, JWT secret, Cloudinary credentials, and Gemini API key

3

Installation & Execution

1. Start backend server
2. Launch AI service
3. Run frontend development server

Seed Data & Deployment

Populating the System

Run `node seedShelters.js` in the server directory to populate the application with default shelters and pets.

Initial admin creation is available via the internal UI path at `/auth/register-shelter-admin`.

Key Endpoints

- API: Express server on port 5000
- AI Worker: FastAPI on port 8000
- Frontend: Vite dev server on port 3000



Thank You